



॥ त्वं ज्ञानमयो विज्ञानमयोऽसि ॥

भारतीय प्रौद्योगिकी संस्थान जोधपुर

एनएच 65, सुरपुरा बाईपास रोड, करवड़, राजस्थान 342037

Indian Institute of Technology Jodhpur

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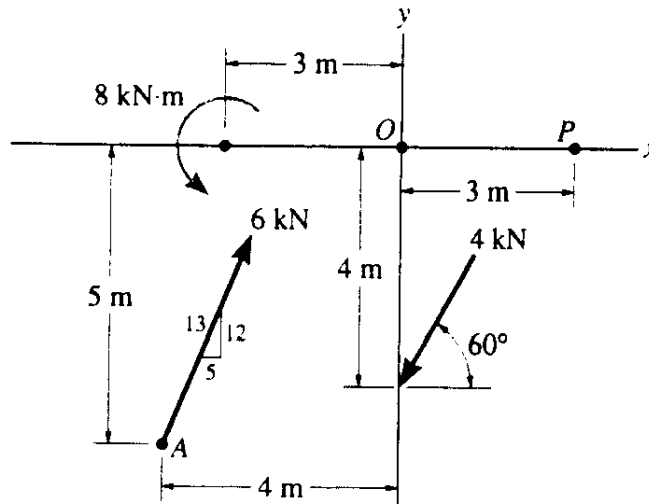
Course: Engineering Mechanics

Code: MEL1010

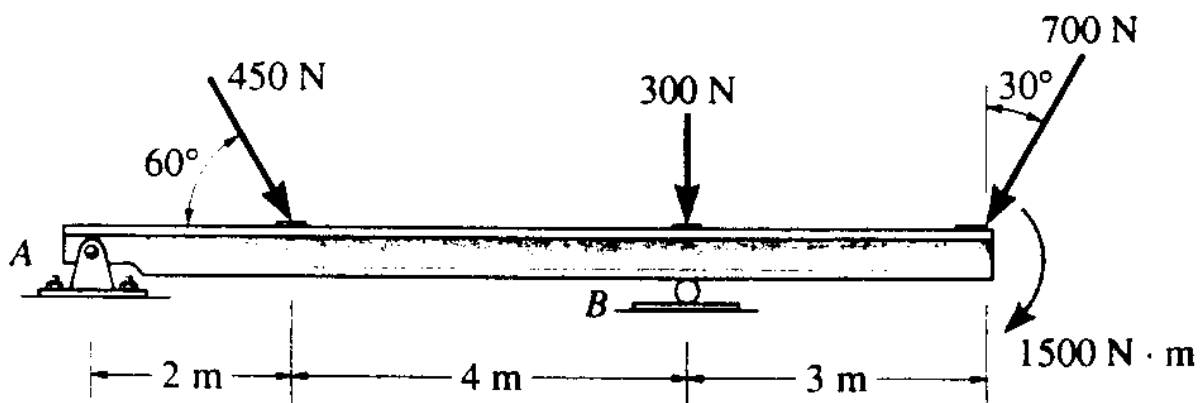
Date: 29/08/2024

Tutorial: 03

Question 01: Replace the force and couple system by an equivalent force and couple moment at point O.



Question 02: Replace the three forces acting on the beam by a single resultant force. Specify where the force acts, Measured from end A.





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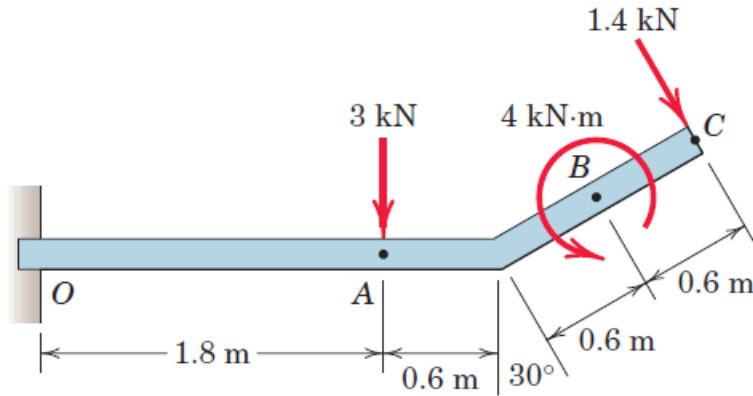
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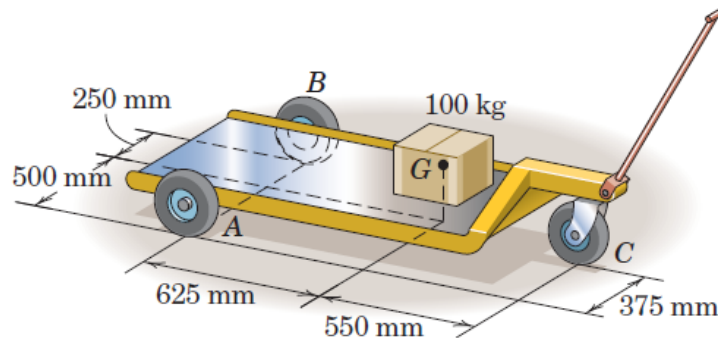
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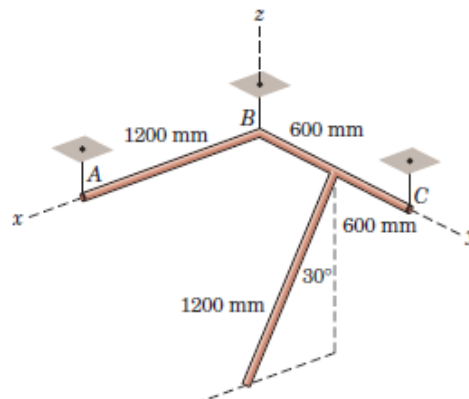
Question 03: The uniform beam has a mass of 50 kg per meter of length. Compute the reactions at the support O. The force loads shown lie in a vertical plane.



Question 04: The three-wheel truck is used to carry the 100-kg box as shown. Calculate the changes in the normal force reactions at the three wheels due to the weight of the box.



Question 05: Each of the three uniform 1200-mm bars has a mass of 20 kg. The bars are welded together into the configuration shown and suspended by three vertical wires. Bars AB and BC lie in the horizontal x-y plane, and the third bar lies in a plane parallel to the x-z plane. Compute the tension in each wire.





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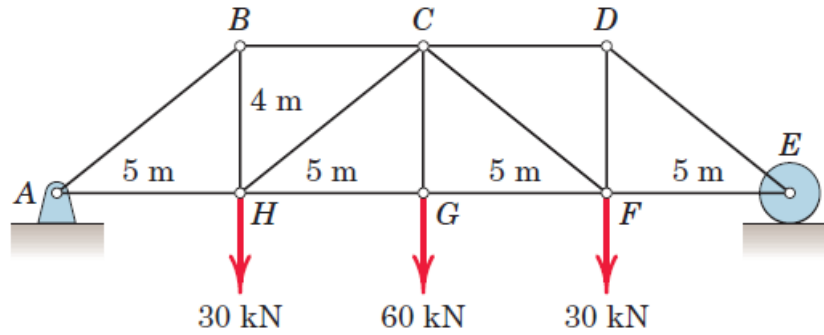
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Question 06: Determine the force in each member of the loaded truss. Make use of the symmetry of the truss and of the loading.



Question 07: A drawbridge is being raised by a cable EI . The four joint loadings shown result from the weight of the roadway. Determine the forces in members EF , DE , DF , CD , and FG .

